MIL-C-20704C
7 February 1980
SUPERSEDING
MIL-C-20704B
28 July 1970

MILITARY SPECIFICATION

CUTTING EDGES, MOLDBOARD, GRADER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cutting edges for grader moldboards.

1.2 Classification. The cutting edges shall be of the following sizes as specified (see 6.2):

5-foot (MS35465-1)
6-foot (MS35465-2)
7-foot (MS35465-3)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation-Packaging Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 3805

Beneficial comments (recommendations, additions deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, Virginia 22060 by using the self-addressed Standarization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-130	- Identification Marking of US Military Property.
MS35465	- Cutting Edge, Moldboard, Grader.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 112 - Methods for estimating the average grain size of metals.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. Unless otherwise specified (see 6.2), the 5-, 6-, and 7-foot size cutting edges shall be fabricated in accordance with MS35465, part number 1, 2, and 3 respectively.

3.2 Material. Material shall be electric furnance, basic open hearth or basic oxygen steel. The grain size of the steel shall conform to ASTM E 112,Size 5 to 8. The steel shall be appreciably free from nonmetallic inclusions, segregations, cracks, seams, pipes, caps, and pits (see 6.3).

3.3 Chemical composition of steel. Unless otherwise specified (see 6.2), the cutting edge shall be of steel conforming to SAE No. 1090. Chemical composition of the SAE No. 1090 steel shall be in accordance with SAE J403.

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3.4 Hardness of steel. Unless otherwise specified (see 6.2), the finished cutting edge shall have a hardness in the range of 271-344 Brinell, 28-37 Rockwell C, to resist wear and abrasion.

3.5 Fasteners. Bolts, nuts, and washers shall be furnished, and shall be packed in a separate container.

3.6 Finish. The cutting edges shall be finished in accordance with the contractor's standard practice for this type of cutting edge.

3.7 Identification marking. The cutting edges shall be marked in accordance with MIL-STD-130.

3.8 Workmanship. Finished cutting edges shall be free from sharp burrs, fins, loose scale, rust and other defects which may impair durability or serviceability. Boltholes shall be finished clean, without burrs caused by punching or cutting.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. All cutting edges of the same type presented at one time shall be considered a lot for the purpose of inspection and acceptance.

4.3 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105, Inspection Level II. AQL shall be 2.5 percent defective.

4.4 Inspection. Each of the sample cutting edges selected in accordance with 4.3 shall be inspected to verify compliance with this specification. Cutting edges in the sample containing one or more defects shall be rejected, and if the number of defective cutting edges in any sample exceeds the acceptance number for that sample, the lot represented by the sample sahll be rejected. Sample cutting edges shall be examined to determine compliance with the requirements listed in table I.

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TABLE I. Examination schedule

Number	Characteristics	Requirment Paragraph
101	Cutting edge size not as specified.	1.2
102	Cutting edge does not conform to description as specified.	3.1
103	Material not as specified.	3.2
104	Fasteners not as specified.	3.5
105	Finish not as specified.	3.6
106	Identification marking not as specified.	3.7
107	Workmanship not as specified.	3.8

4.5 Certification.

4.5.1 Chemical composition. The contractor shall submit proof satisfactory to the procuring agency that the cutting edges to be supplied under this specification meet the chemical composition and requirements specified in 3.3 or as stated in the invitation for bid or contract.

4.5.2 Hardness of steel. The contractor shall submit proof satisfactory to the procuring agency that the cutting edges to be supplied under this specification meet the hardness requirement specified in 3.4 or as stated in the invitation for bid or contract.

4.6 Inspection of packaging. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product. Sampling for examination shall be in accordance with MIL-STD-105. Samples shall be examined for the characteristics listed in table II. AQL shall be 2.5 percent defective.

TABLE II. Examination schedule - packaging

Number	Characteristics	Requirement Paragraph
108	Preservation not in accordance with level A as specified.	5.1.1
109	Commercial preservation not in accordance with MIL-STD-1188 as specified.	5.1.2
110	Packing not in accordance with Level A as specified.	5.2.1
111	Commercial packing not in accordance with MIL-STD-1188 as specified.	5.2.2
112	Marking not in accordance with MIL-STD-129 for level A.	5.3
113	Commercial marking not in accordance with MIL-STD-1188 as specified.	5.3

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Cutting edges shall be preserved in accordance with MIL-P-116, Method I, applying Type P-1 preservation to all unpainted base metal surfaces of the cutting edges, bolts, and nuts.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188, except that unit container is not required.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Cutting edges in quantities of two shall be bolted together to form a uniform bundle or pack with two standard plow bolts, or with the size and type required for attaching the cutting edges to the moldboard. Strapping may be used as an alternate to band the cutting edges together.

5.2.2 Commercial. Commercial packing shall be in accordance with MIL-STD-1188, except that a shipping container is not required.

5.3 Marking. In addition to any special marking required, the bundles shall be marked in accordance with MIL-STD-129 for level A. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Cutting edges covered by this specification are intended for use on grader moldboards.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title , number and date of this specification.
(b) Size cutting edge required (see 1.2).
(c) When fabrication other than specified is required (see 3.1).
(d) When chemical composition other than specified is required (see 3.3).
(e) When hardness other than specified is required (see 3.4).
(f) Degree of preservation and degree of packing required (see 5.1 and 5.2).

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6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

Custodians:
Army - ME

Review activity:
Army - AT
DLA - CS

User activities:
Army - CE
Navy - YD

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-20704C Cutting Edges, Moldboard, Grader

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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